Parametrized Text in KiCad

In many PCB designs, adding custom text to the silkscreen layer is a common requirement. Examples include marking the project title, version number, or board revision.

But what if this information changes frequently or appears in multiple places across your design?

Manually updating every text label can quickly become tedious. Furthermore, if you use tools such as Kibuzzard to generate text as footprints, these footprints may be deleted during an **Update PCB from Schematic** operation if **Delete footprints with no symbols** is enabled.

Solution

Use parametrized silkscreen text with non-electrical symbols and custom text variables. This way, your text becomes dynamic and automatically updates according to the project settings.

What You'll Learn

In this tutorial, we will walk through how to:

- Create a custom text variable in your KiCad project.
- Design a non-electrical symbol that references this variable.
- Associate the symbol with a custom footprint that displays the text on the silkscreen layer.

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1. Create a Text Variable

KiCad allows you to define custom text variables that can be reused throughout your project.

Steps

1. Open your KiCad project.
2. Open either the Schematic Editor or PCB Editor.
3. In the Schematic Editor, select **File → Schematic Setup**. In the PCB Editor, select **File → Board Setup**.
4. Navigate to **Text & Graphics → Text Variables**.
5. Click **Add** and define a new variable.
6. Set the following values:

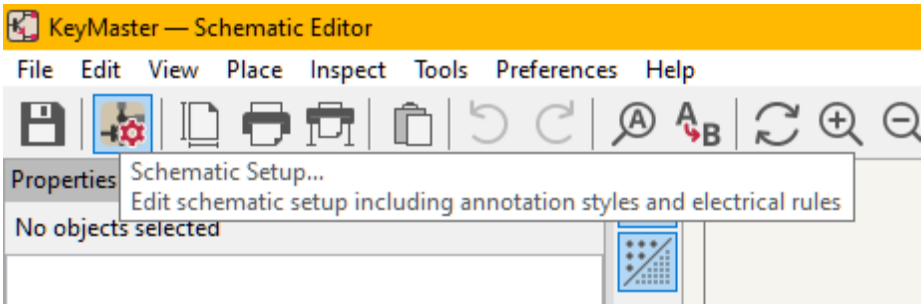


Figure 1: Schematic Setup

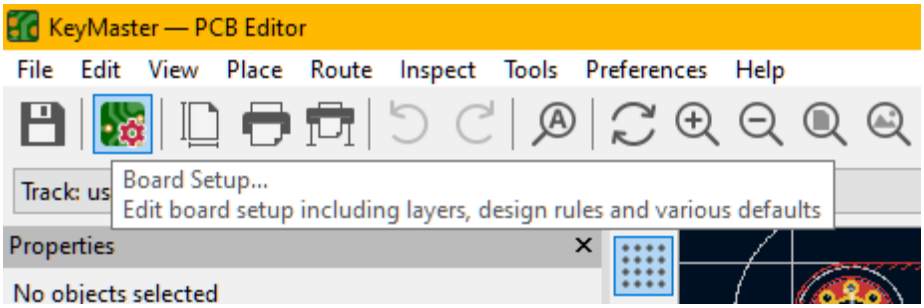


Figure 2: Board Setup

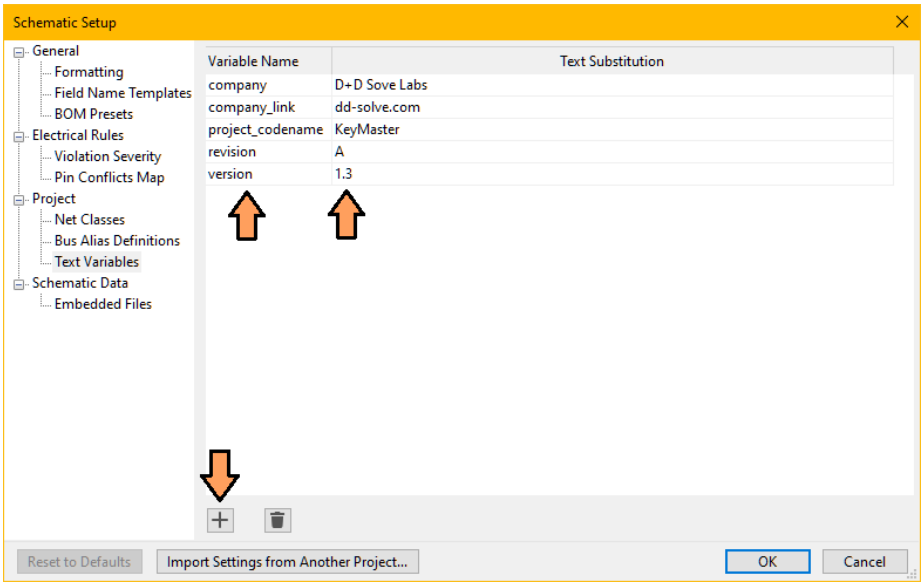


Figure 3: Text Variables

- **Variable Name:** `version`
- **Text Substitution:** `1.3`

7. Click **OK** to save.

   Note: You can now use `${version}` anywhere in your project to reference this value.

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2. Create a Symbol to Hold the Text

Next, we will create a dummy symbol that references our text variable. This allows us to place the text in the schematic and preserve its association with the PCB footprint during schematic updates.

Steps

1. Open **Symbol Editor** (Tools → Symbol Editor).

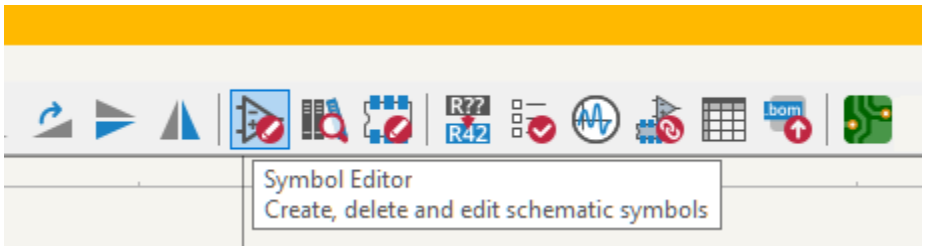


Figure 4: Symbol Editor

2. Create a new library or add an existing one.
3. Choose whether the library should be **Global** or **Project** specific.
4. Name the symbol library, for example, `Text_Label.kicad_sym`.
5. Click **New Symbol**.
6. Name the symbol, for example, `text_label`.
7. Assign a reference designator, for example, `TXT`.
8. Open **Symbol Properties** from the menu bar.
9. Configure the symbol properties:
   - Set the symbol value (it is not particularly important at this stage).
   - Enable **Exclude from simulation** (optional).
   - Enable **Exclude from Bill of Materials** to avoid including this dummy component in the BOM.
10. Save the symbol, keeping the Symbol Editor open.

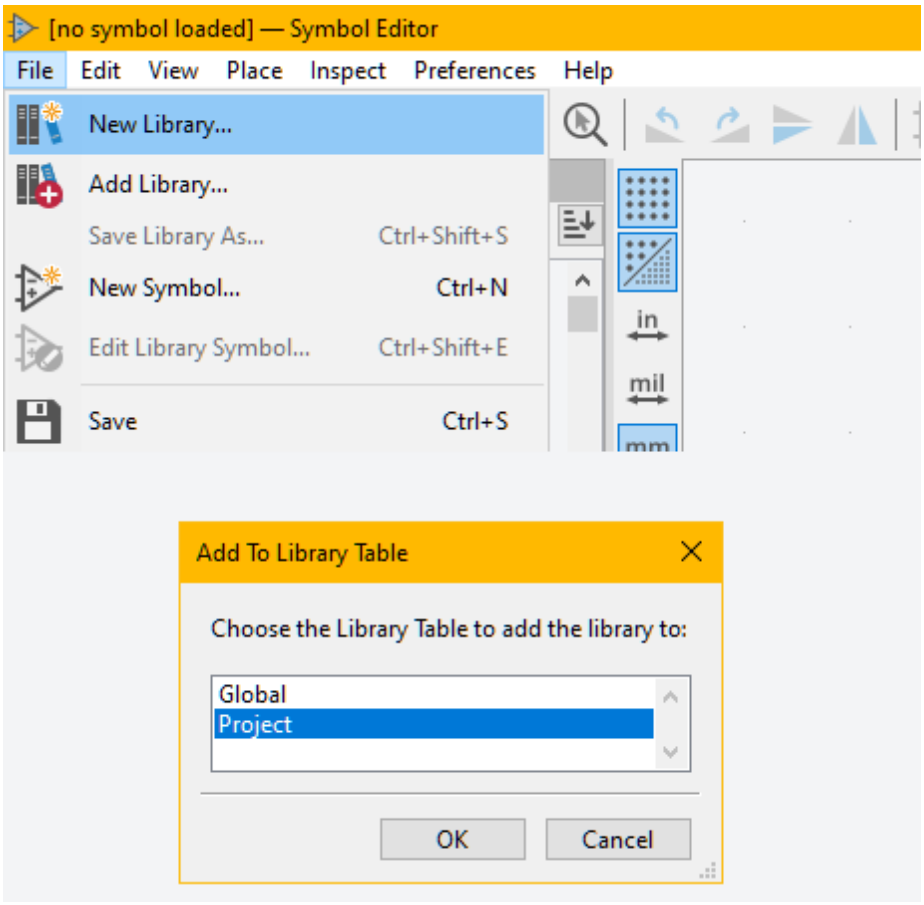


Figure 5: New Library

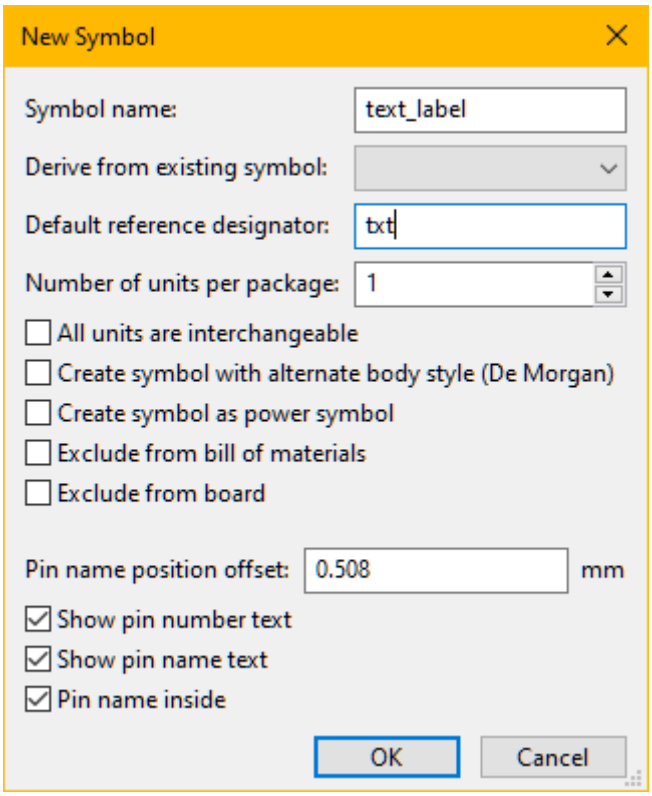


Figure 6: New Symbol

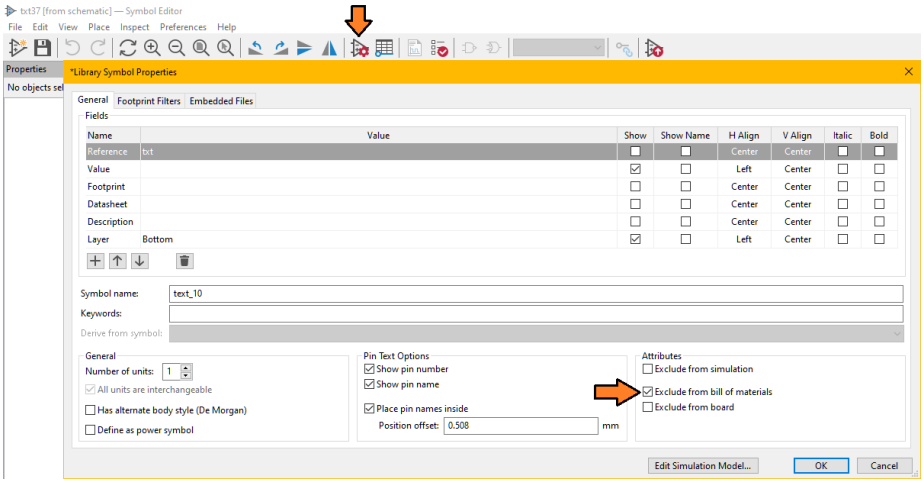


Figure 7: Symbol Properties

Note: This symbol acts purely as a label and does not affect the electrical functionality of the circuit.

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3. Create a Footprint for the Silkscreen

Finally, we will associate the symbol with a footprint that displays the text on the PCB silkscreen.

Steps

1. Open **Footprint Editor**.

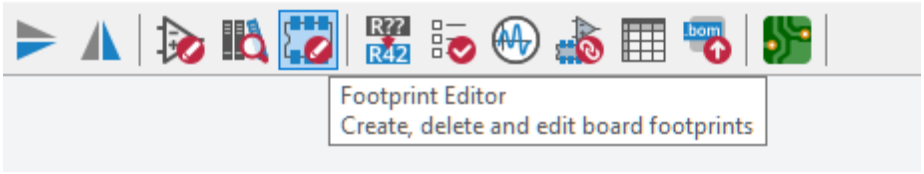


Figure 8: Open Footprint Editor

2. Create a new library or add an existing one.
3. Choose whether the library should be **Global** or **Project** specific.
4. Name the footprint library, for example, `text_label.pretty`.
5. Click **New Footprint** (File → New Footprint).
6. Name the footprint, for example, `txt_label`, and save it in the library you just created (`text_label.pretty`).
7. Customize the **Footprint Properties**, particularly the text settings:
   - Font and text size.
   - Horizontal and vertical alignment.
   - Text knockout settings, if required.
8. Most importantly, select the appropriate layer:
   - `F.Silkscreen` for front silkscreen.
   - `B.Silkscreen` for back silkscreen.
9. Save the footprint.

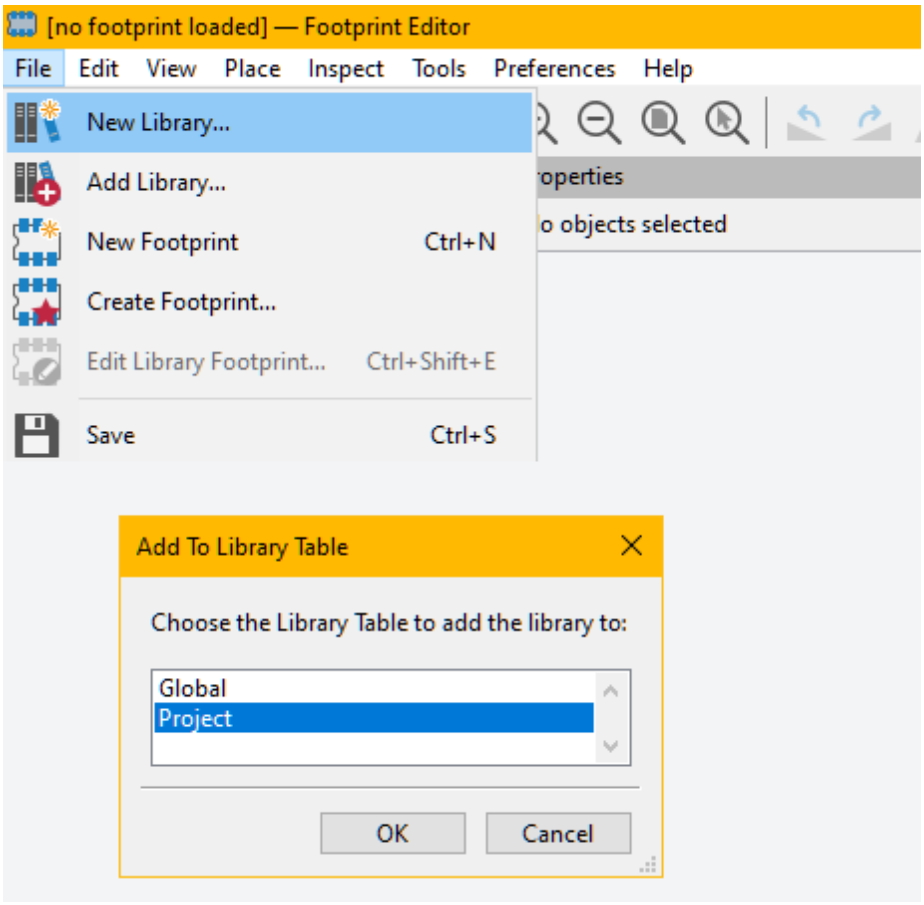


Figure 9: New Footprint Library

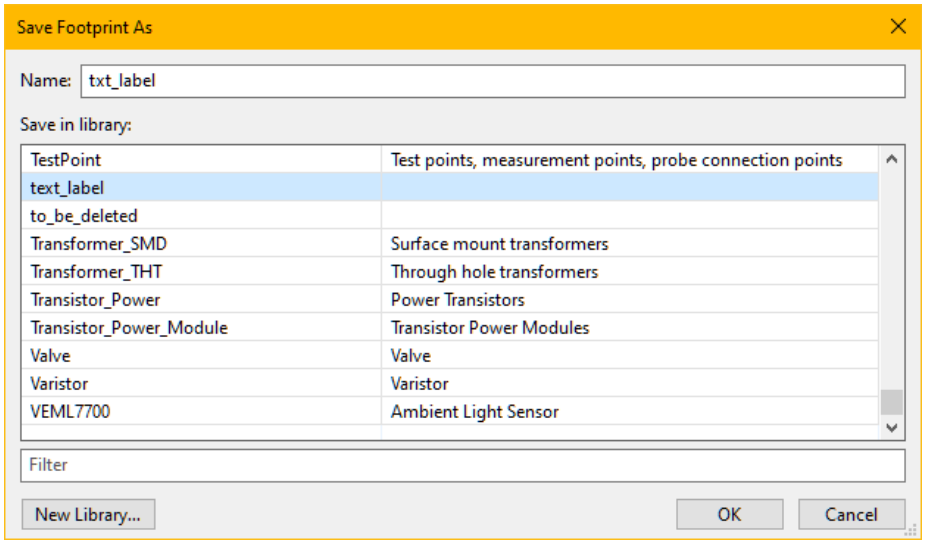


Figure 10: New Footprint

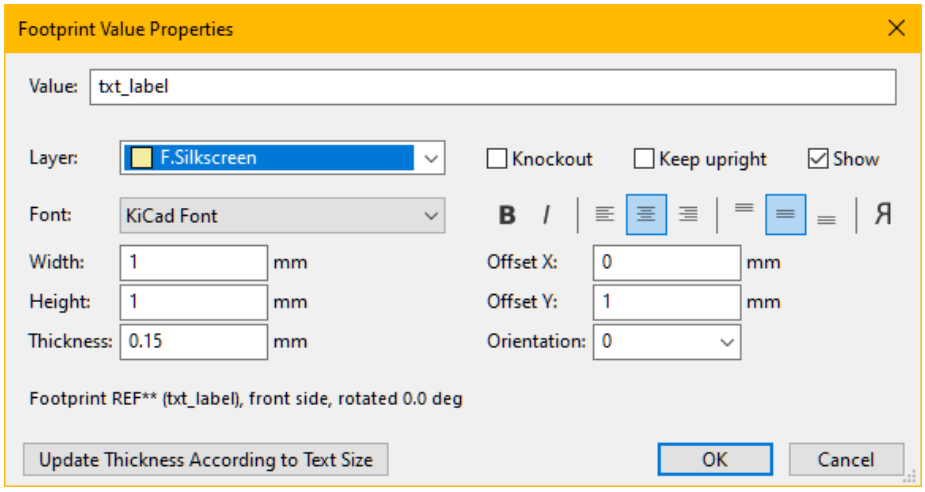


Figure 11: Footprint Value Properties

4. Link the Symbol to the Footprint

Now we need to associate the symbol created earlier with the new footprint.

1. Return to the **Symbol Editor**.
2. Open **Symbol Properties**.

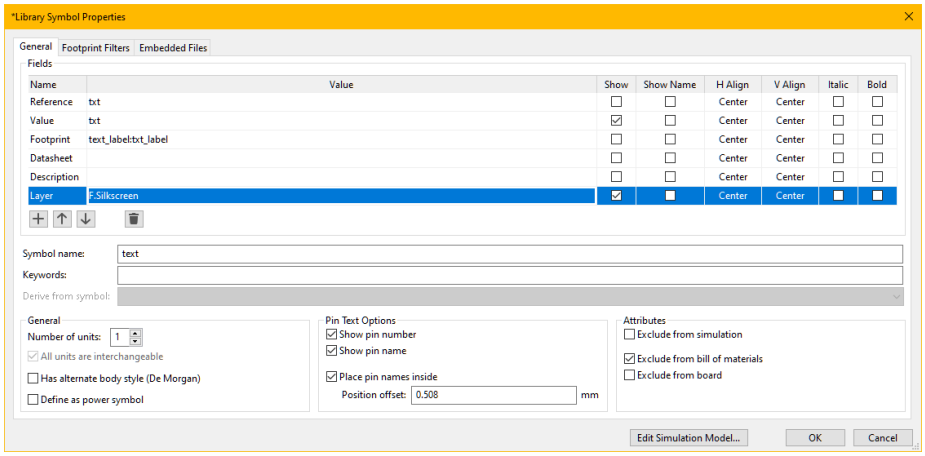


Figure 12: Footprint Assignment

3. Set the footprint to the one you just created: `text_label:txt_label`.
4. Save all changes and return to the Schematic Editor.
5. Place the `text_label` symbol in your schematic.
6. Set its value to `${version}`.
7. Run **Update PCB from Schematic** to synchronize the PCB with the schematic.

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5. Final Result

Your PCB will now display the version number on the silkscreen, dynamically controlled by the project text variable.

If you need to change the board version, simply update the `version` variable in the project settings. All instances referencing `${version}` will automatically reflect the new value after the appropriate PCB update.

This approach is particularly useful for: - PCB revision numbers. - Project names and identifiers. - Hardware version labels. - Other information that may change during development.

Download

You can download the example KiCad library here:

Text Label – KiCad Library